

SUNFLOWER

Solar Tracker for acquiring maximum power



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Efficiency Optimization with Single-Axis Solar Tracking

ROAD MAP

1 REQUIREMENTS

- Arduino UNO
- LDR Sensors (2)
- SG90 Servo Motor (1)
- Small Solar Panel
- 10k Resistors (2)
- Breadboard
- Jumper Wires
- USB Cable/Power

2 SYSTEM SETUP

- Assemble all components
- Connect LDRs in array
- Mount solar panel on servo stand
- Power up Arduino

3 SENSOR CALIBRATION

- Read both LDR values
- Set threshold values
- Test in different light conditions

4 TRACKING ALGORITHM

- Calculate LDR difference
- Determine direction
- Control servo motor

5 TESTING & TUNING

- Test tracking accuracy
- Adjust thresholds
- Improve response

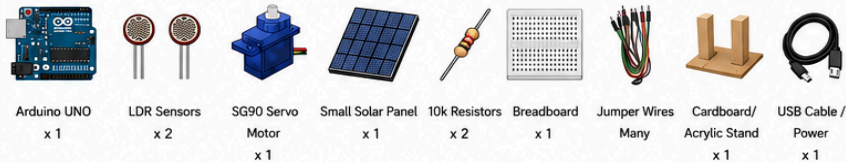
6 OPTIMIZE & DEPLOY

- Optimize for efficiency
- Final testing
- Deploy system

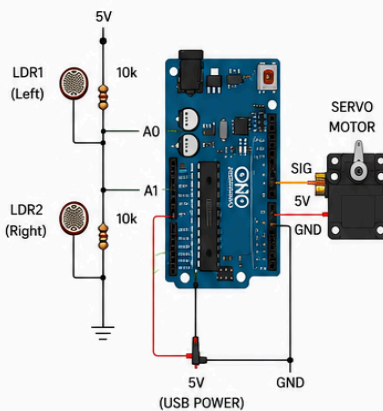
7 RESULT & MONITOR

- Continuous tracking
- Max power output
- Monitor performance

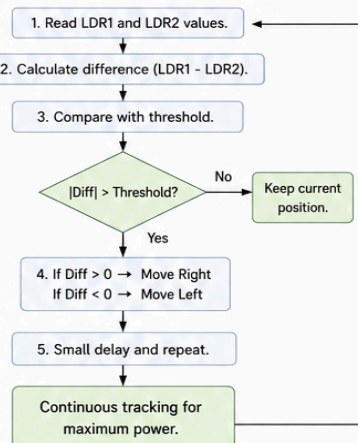
COMPONENTS USED



CIRCUIT SCHEMATIC



WORKING PRINCIPLE & FLOW



SENSOR DATA (EXAMPLE)

LDR VALUES (Sample)		
Condition	LDR1 (Left)	LDR2 (Right)
Sun Left	230	120
Sun Center	180	175
Sun Right	110	240

DIFFERENCE (LDR1 - LDR2)		
Positive	→	Move Right
Negative	→	Move Left
Around 0	→	Stay

SOFTWARE

- Arduino IDE
- Libraries: None
- Key Logic:
 - Read Analog Values
 - Calculate Difference
 - Compare Threshold
 - Control Servo



DEMO FLOW

1. Power On
2. System Init
3. Read LDRs
4. Auto-Tracking Active
5. Track Sun Continuously
6. Maximize Power Output



APPLICATIONS

- Remote Solar Stations
- Rooftop Off-Grid Systems
- Efficient Charging Hubs
- Research & Education
- Solar Powered Devices



Track the sun. Maximize the power.

Aim:

To develop an automatic solar tracking system that continuously tracks sunlight using LDR sensors and adjusts the solar panel position using a servo motor to maximize solar energy absorption.

Objectives:

- Detect sunlight intensity using two LDR sensors.
- Compare the light falling on both LDRs.
- Control the servo motor using Arduino UNO.
- Automatically rotate the solar panel towards the direction of maximum sunlight.
- Improve the efficiency of solar power generation.

Methodology:

The system uses two LDR (Light Dependent Resistor) sensors placed on either side of a solar panel. The LDRs detect the intensity of sunlight from different directions and send the values to the Arduino UNO. The Arduino continuously compares the readings from both sensors. If one LDR receives more light than the other, the Arduino drives the servo motor to rotate the solar panel towards the brighter side until the difference between the LDR sensor values is less than 30 units. This process continues automatically throughout the day, ensuring that the solar panel remains aligned with the maximum available sunlight.

Results and Discussion:

The developed solar tracking system successfully tracked the direction of maximum sunlight in real time. The LDR sensors accurately detected variations in light intensity, and the Arduino UNO processed the sensor readings effectively. The servo motor responded quickly and adjusted the panel position whenever a difference in light intensity was detected. As a result, the solar panel remained oriented towards the strongest light source, demonstrating improved sunlight utilization compared to a fixed-position panel. The system proved to be simple, cost-effective, and suitable for small-scale solar energy applications.

Conclusion:

The Sunflower solar tracking system was successfully designed and implemented using an Arduino UNO, LDR sensors, and a servo motor. The system effectively tracked sunlight and automatically adjusted the solar panel's orientation to maximize light exposure, demonstrating a clear practical application of embedded systems in renewable energy optimization.

Empirical testing confirmed the system's efficacy:

- **Sensor Sensitivity:** The potential difference across the LDRs varied from 1.7 V (in ambient/low-light conditions) to 2.1 V (under direct illumination), providing a distinct threshold for the Arduino to trigger motor adjustments.
- **Power Optimization:** The solar panel demonstrated a measurable increase in output, generating 3.2 V under direct light compared to 2.2 V in shaded conditions.

These results validate that the dynamic alignment provided by Sunflower significantly enhances energy harvesting efficiency compared to static, fixed-position solar panels.

Future Scope:

- Implement dual-axis tracking for higher accuracy.
- Integrate IoT technology for remote monitoring and control.
- Add battery charging and power management systems.
- Use more efficient sensors and motors for improved performance.
- Apply the system to large-scale solar power installations.

